STAT 5200, Notes 5[1]

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Readings. Read Section 12.8 in W, which deals with simulation and resampling. In these notes I'll discuss the use of simulation to determine standard errors and introduce bootstrapping. I'll consider the parametric bootstrap, the nonparametric bootstrap, the nonparametric residual bootstrap for regression, and the nonparametric bootstrap for regression; and also briefly discuss the jackknife. In the companion notes for the lab session, I will use R to calculate standard errors in examples via the bootstrap and the jackknife.

Background and motivation. In an introductory statistics course one learns about random sampling from some population or some distribution, with the purpose of estimating the parameter that characterizes the population or distribution, and perhaps testing hypotheses about the parameter. One assumes there is a sample of observations, $x_1, ..., x_N$ drawn independently from a distribution with parameter θ. For example, suppose the sample is drawn from the normal distribution with mean μ and variance σ^2, and one estimates these parameters by the sample mean and sample variance,

$$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \quad s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2.$$

The properties of these two estimators are known analytically for any sample size: $\overline{x}$ is normal with mean μ and variance σ^2/N, and $(N-1)s^2/\sigma^2$ has a chi-square distribution with $N-1$ degrees of freedom, so that the estimator s^2 has mean σ^2 and variance $2\sigma^4/(N-1)$. Moreover, the estimators $\overline{x}$ and s^2 are independent. In the classical linear model setting with fixed values of the explanatory variables, the situation is similar. That is, for any sample size N, exact results can be given for the joint distribution of the least squares estimator of the parameter vector β and the estimator of the parameter σ^2, which is the variance of the disturbance term (assuming homoscedasticity). This allows one to perform statistical inference (calculation of standard errors, construction of confidence intervals, and determination of p-values for hypothesis tests) very cleanly and easily. Of course, the assumptions in the classical linear model that the explanatory variable values are fixed, and that the errors are independently normally distributed with constant variance, form an ideal framework that one does not expect to hold exactly in a real setting where data have been observed. Rather, one behaves as if the assumptions are approximately valid in the real setting. Further, one needs to

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use residual diagnostics to examine the validity of the assumptions. If the assumptions appear to be violated, given the residual diagnostics, it may be possible to make some adjustments, such as transforming the data, modifying or removing outlier values, and employing weighted least squares or calculating HC standard errors.

In the regression analysis we have been considering in this course, the situation is different. The explanatory variables are assumed to be random, and exact distributional results cannot be given in finite samples for the estimators of the parameters. One uses asymptotic distributions of the estimators to perform statistical inference. One hopes the asymptotic distributions give decent approximations for the properties of the estimators for the finite sample size one is working with.

Sometimes one has an estimator $\widehat{\theta}$ for a parameter θ, and one wants to estimate the parameter $\mathbf{g}(\theta)$, where $\mathbf{g}(\cdot)$ is some transformation. The natural estimator is the plug-in estimator $\mathbf{g}(\widehat{\theta})$, and one can try to use the delta method to determine the standard errors and the distribution for this estimator. The results obtained from the delta method are asymptotic approximations, and use of the delta method depends on having usable expressions for the standard errors and the distribution of $\widehat{\theta}$ itself (typically they are themselves asymptotic approximations). And, of course, the delta method requires smooth transformations $\mathbf{g}(\theta)$ for which derivatives can be calculated and are well-behaved.

There are still situations where one does not even have asymptotic distributional results which can be employed to perform statistical inference, given the estimators one has constructed – the estimators are complicated, and/or it is difficult to get an analytical grip on them. Furthermore, some regression settings involve the presence of heteroscedasticity, and this further complicates matters.

Even when one can obtain asymptotic approximations, perhaps by the delta method, one is still concerned about the accuracy and/or relevance of the approximations. It is in such situations that one can often usefully employ simulation and resampling methods.

Monte Carlo simulation. A Monte Carlo study uses repeated sampling and can sometimes be used to obtain the statistical properties of an estimator $\widehat{\theta}$ of a parameter θ, for example, $E[\widehat{\theta}]$ or $Var(\widehat{\theta})$. The parameter characterizes some population distribution. To proceed, one chooses a specific value of the parameter for the population distribution, say θ_0. Then one draws a random sample of size N from this specific population distribution and constructs the estimate $\widehat{\theta}^{(1)}$ of θ_0 from the sample. One repeats this sampling M times, resulting in the M estimates $\widehat{\theta}^{(m)}$, $m = 1, ..., M$, all estimates of θ_0. The sample average and the sample variance of these M estimates then are estimates of $E[\widehat{\theta}]$ and $Var(\widehat{\theta})$, respectively. One can consider other properties of the estimator $\widehat{\theta}$, and also an arbitrary parametric function $\mathbf{g}(\theta)$ and its estimator $\mathbf{g}(\widehat{\theta})$. A thorough Monte Carlo study varies the values of θ_0, N and M. This hopefully gives one a representative view of the landscape describing the statistical properties of the estimator. Wooldridge's discussion in the last three paragraphs of Section 12.8 is very good. One of the points he emphasizes is that many econometric estimation methods use an asymptotic distribution for the estimators and do not assume the form of the specific underlying distribution which generated the data. Thus, use of Monte Carlo simulation to get an understanding of the properties of the estimators may not be feasible.

And, in many situations, the usefulness of a Monte Carlo simulation is questionable. Typically we have a single sample assumed to come from some population, and we don't know the value θ_0. So we really don't know whether a Monte Carlo study is applicable to the population from which our sample is assumed to be drawn. Still, the Monte Carlo study can often be helpful in providing some guidance on adjustment of standard errors and p-values.

The bootstrap for a location parameter. Invention of the bootstrap is attributed to Bradley Efron (1979, Bootstrap methods: Another look at the jackknife, *The Annals of Statistics* **7** 1-26). Subsampling and resampling methods had been advocated earlier by others, including the economist Julian Simon. For information about Simon, see http://en.wikipedia.org/wiki/Julian_Simon. The article by Peter Hall (2003, A short prehistory of the bootstrap, *Statistical Science* **18** 158-167) discusses forerunners of the bootstrap and includes comments about Simon and his resampling methods; I've posted it on Canvas.

To understand the bootstrap, let's reexamine the sampling setting and define the empirical distribution function.

The empirical distribution. Assume we collect a sample of N independent observations $x_1, ..., x_N$, all from the distribution with cumulative distribution function (cdf) F. Given the sample, the *empirical distribution function* $\widehat{F}_N(x)$ is the cdf which has a jump of magnitude $1/N$ at each observed sample value. The empirical distribution function is an estimate of the cdf F. By the Glivenko–Cantelli theorem, $\widehat{F}_N(x)$ converges to $F(x)$, with probability 1, uniformly in x as N tends to infinity. We can define $\widehat{F}_N(x)$ by

$$\widehat{F}_N(x) = \#(x_i \leq x)/N, \quad -\infty < x < \infty,$$

where $\#(x_i \leq x)$ is the count of the number of sample values less than or equal to x.

The empirical distribution is discrete, with probability $1/N$ attached to each of the data values x_i. Thus the mean of the empirical distribution is

$$\int_{-\infty}^{\infty} x d\widehat{F}_N(x) = \frac{1}{N}\sum_{i=1}^{N} x_i = \overline{x},$$

the sample average, and the variance of the empirical distribution is

$$\int_{-\infty}^{\infty} (x - \overline{x})^2 d\widehat{F}_N(x) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \overline{x})^2.$$

Note the division by N, rather than $N - 1$.

Estimation of the variance of a statistic – motivation. Assume the distribution from which we have sampled is characterized by a parameter θ, which we estimate by $\widehat{\theta}$. Furthermore, we want to estimate $g(\theta)$, and we do so with $g(\widehat{\theta})$. Here, I am assuming $g(\cdot)$ is a scalar. How do we determine the standard error of $g(\widehat{\theta})$ if we don't have an analytic expression for the variance of $\widehat{\theta}$; and if we can't use the delta method because the calculation of the derivative of g is too complicated, or because g is not smooth and doesn't have a derivative, or because we are worried that the approximate calculation provided by the delta method will not be sufficiently accurate? The bootstrap provides a method of determining the standard error of $g(\widehat{\theta})$.

The nonparametric bootstrap procedure. Suppose we have a single sample of values, $x_1, ..., x_N$ available. The motivation behind the bootstrap is that $\widehat{F}_N(x)$ is an estimate of $F(x)$. So we take repeated samples from the empirical distribution and in each sample calculate the estimate $g(\widehat{\theta})$. To determine the standard error of $g(\widehat{\theta})$, we calculate the sample variance of these estimates. The procedure is as follows:

(a) Select a sample of size N with replacement from the data $x_1, ..., x_N$. Calculate $g(\widehat{\theta}^{(1)})$ from this sample.

(b) Repeat step (a) $B - 1$ times. Now one has B estimates,

$$g(\widehat{\theta}^{(b)}), \quad b = 1, ..., B. \tag{1}$$

(c) Calculate the sample variance of the B estimates of $g(\theta)$,

$$\widehat{Var}_{boot} = \frac{1}{B-1} \sum_{b=1}^{B} \left(g(\widehat{\theta}^{(b)}) - \frac{1}{B} \sum_{j=1}^{B} g(\widehat{\theta}^{(j)}) \right)^2. \tag{2}$$

This is the bootstrap estimate of the variance of $g(\widehat{\theta})$. The bootstrap standard error of $g(\widehat{\theta})$ is then se_{boot}, calculated as the square root of $\widehat{Var}_{boot}$.

If the N data values $x_1, ..., x_N$ are distinct, the number of distinct bootstrap samples is

$$\binom{2N-1}{N}.$$

The bootstrap procedure can be used to estimate other properties of the estimator $g(\widehat{\theta})$, such as its median, and even its distribution. For the former, just calculate the median of the values (1). For the latter, form a histogram or a quantile-quantile plot from the values (1).

Note that the nonparametric bootstrap procedure involves two approximations. The first is the approximation of the underlying cdf F by the empirical distribution function $\widehat{F}_N$ (recall the statement of the Glivenko-Cantelli theorem). The second approximation is illustrated by the approximation of the variance of $g(\widehat{\theta})$ by formula (3) when the "true distribution" is $\widehat{F}_N$.

In performing the bootstrap, one does not have to use sample size N in selecting bootstrap samples; it is common to do so, though. One also has to select B, and this is usually chosen to be large (1,000, e.g.).

There are several ways to calculate bootstrap confidence intervals. Here are two:

(a) This uses the normal distribution, assuming it is applicable. The interval is

$$g(\widehat{\theta}) \pm z_{\alpha/2} se_{boot},$$

where $g(\widehat{\theta})$ is the calculation of the estimate from the original sample.

(b) Take the endpoints of the $100(1-\alpha)$ per cent confidence interval to be the $\alpha/2$ and $1-\alpha/2$ quantiles of the distribution of the B bootstrap estimates of $g(\theta)$.

The parametric bootstrap. The parametric bootstrap resembles a Monte Carlo simulation. Denote the cdf of the distribution from which we have drawn the sample by $F(x;\theta)$. We require knowledge of the functional form of F for this to work. Construct $\widehat{\theta}$ from the data sample. Then we select B random samples, each from $F(x;\widehat{\theta})$. In each of these samples we calculate an estimate of $g(\theta)$, giving us $g(\widehat{\theta}^{(b)})$, $b = 1, ..., B$. Then, to obtain the variance of $g(\widehat{\theta})$, for example, perform step (c) on page 4.

The bootstrap for regression.

The nonparametric residual bootstrap for regression. This is described in W. One fits a regression to the data and calculates the residuals. Then one takes bootstrap samples *of the residuals*, and for each bootstrap sample one uses the estimated regression function to construct values of the response variable. Thus, one obtains B samples of the response.

Assume the model is $y = \mathbf{x}\beta + u$, that u is independent of $\mathbf{x}$, and that we have a sample $(\mathbf{x}_i, y_i)$, $i = 1, ..., N$. Estimate β by $\widehat{\beta}$ (for example, use least squares) and form the residuals $\widehat{u}_i = y_i - \mathbf{x}_i\widehat{\beta}$, $i = 1, ..., N$. Now form B bootstrap samples from these residuals (that is, to obtain each of the bootstrap samples we sample with replacement from the empirical distribution of the residuals). From the bth bootstrap sample, the residuals are $\widehat{u}_i^{(b)}$, and we form the values

$$y_i^{(b)} = \mathbf{x}_i\widehat{\beta} + \widehat{u}_i^{(b)}, \quad i = 1, ..., N.$$

Now we have B bootstrap data sets

$$\{(\mathbf{x}_i, y_i^{(b)}), \quad i = 1, ..., N\}, \quad b = 1, ..., B.$$

From these data sets calculate $\widehat{\beta}^{(b)}$, $b = 1, ..., B$. Then use a calculation such as (3) to obtain variances and covariances for the estimates of the elements of the parameter vector, and thus standard errors for the estimates.

The wild (residual) bootstrap for regression. The above method (nonparametric residual bootstrap) samples the empirical distribution determined by the residual values. Hence it assumes that there is a single underlying distribution for the disturbance term in the model, that is, it assumes homoscedasticity. Wooldridge comments that the *wild bootstrap* can be used to obtain consistent standard errors when heteroscedasticity prevails.

The wild bootstrap is a slightly modified version of the residual bootstrap:

From the bth bootstrap sample, we form the values

$$y_i^{(b)} = \mathbf{x}_i\widehat{\beta} + v_i^{(b)}\widehat{u}_i, \quad i = 1, ..., N.$$

where $v_i^{(b)}$ is an independent random variable (mean zero, variance one). The popular choice for the distribution of $v_i^{(b)}$ is

$$v_i = \begin{cases} 1 \text{ with probability } 1/2 \\ -1 \text{ with probability } 1/2 \end{cases}$$

or $v_i \sim N(0,1)$. The former seems to be preferred because it introduces minimal randomness while keeping the heteroscedasticity.[3]

Now we have B bootstrap data sets

$$\{(\mathbf{x}_i, y_i^{(b)}), \quad i = 1, ..., N\}, \quad b = 1, ..., B.$$

The nonparametric bootstrap for regression. Instead of resampling the residuals and using the same values of the explanatory variables for each resample outcome, one can employ a bootstrap procedure which resamples the $(\mathbf{x}_i, y_i)$ values. This is implemented by the steps described above in the section titled "The nonparametric bootstrap procedure."

What can we do with bootstrapping? Let's consider some further comments about the bootstrap. To understand the bootstrap, it is instructive to think in terms of "the real world" and "the bootstrap world." The bootstrap world mimics the real world. Let's consider estimation of the bias of an estimator and hypothesis testing via the t-test.

Standard error. Calculate the sample variance of the B estimates of β for a general function $g()$,

$$\widehat{Var}_{boot} = \frac{1}{B-1} \sum_{b=1}^{B} \left(\widehat{\beta}^{(b)} - \frac{1}{B} \sum_{j=1}^{B} \widehat{\beta}^{(j)} \right)^2 . \tag{3}$$

This is the bootstrap estimate of the variance of $\widehat{\beta}$. The bootstrap standard error of $\widehat{\beta}$ is then se_{boot}, calculated as the square root of $\widehat{Var}_{boot}$.

Confidence interval. There are several ways to calculate bootstrap confidence intervals. Here are two:

(a) This uses the normal distribution, assuming it is applicable. The interval is

$$\widehat{\beta} \pm z_{\alpha/2} se_{boot},$$

where $\widehat{\beta}$ is the calculation of the estimate from the original sample.

(b) Take the endpoints of the $100(1-\alpha)$ per cent confidence interval to be the $\alpha/2$ and $1-\alpha/2$ quantiles of the distribution of the B bootstrap estimates of β.

[3] $Var(v_i u_i|x_i) = Var(v_i)Var(u_i|x_i) = Var(u_i|x_i)$ as long as v_i is independent of (u_i, x_i) and has variance one.

Estimation of bias. Suppose we want to estimate a parameter which characterizes some population of interest. In the simplest situation in the real world, we draw a sample $x_1, ..., x_N$ of independent and identically distributed observations from the population. Use the notation β to denote the true value of the parameter, what we are trying to estimate. We construct an estimator $\widehat{\beta}$ of β. Then the bias of the estimator is

$$E[\widehat{\beta}] - \beta, \tag{4}$$

where the expectation is with respect to the population distribution.

In the real world we have only one sample, and thus we cannot estimate the bias. We can, however, use the bootstrap world to estimate the bias. The sample values constitute the population in the bootstrap world, and the population distribution in the bootstrap world is uniform on the sample values. In this bootstrap world we draw B samples of size N, say, with replacement (from the data $x_1, ..., x_N$). In each sample we construct our estimator of β. This yields

$$\widehat{\beta}^{(b)},\ b = 1, ..., B. \tag{5}$$

The bootstrap bias estimator is the analog of (4) in the bootstrap world, or

$$\frac{1}{B}\sum_{b=1}^{B} \widehat{\beta}^{(b)} - \widehat{\beta}. \tag{6}$$

Note that the centering is at the estimate obtained from the real-world sample. It follows that a bias-corrected estimator of β is

$$\widehat{\beta} - \left(\frac{1}{B}\sum_{b=1}^{B} \widehat{\beta}^{(b)} - \widehat{\beta}\right) = 2\widehat{\beta} - \frac{1}{B}\sum_{b=1}^{B} \widehat{\beta}^{(b)}. \tag{7}$$

The histogram of the bootstrap values in (5), which is in the bootstrap world, corresponds to the density of the estimator $\widehat{\beta}$ in the real world.

Hypothesis testing. Consider testing $H_0 : \beta = c$ versus a two-sided alternative. From the real world sample the t-statistic is

$$t = \frac{\widehat{\beta} - c}{se(\widehat{\beta})}, \tag{8}$$

where the denominator houses the standard error of the estimator.

We reject the null hypothesis if the absolute value of t exceeds the $(1 - \alpha/2)$ quantile of the t distribution (with appropriate degrees of freedom), for a test at level α. Alternatively, we use $(1 - \alpha/2)$ quantile of the standard normal distribution.

To apply bootstrap testing, we collect B bootstrap samples, and for each we calculate a t-statistic, yielding

$$t^{(b)} = \frac{\widehat{\theta}^{(b)} - \widehat{\theta}}{se(\widehat{\theta}^{(b)})}, \quad b = 1, ..., B. \tag{9}$$

Note the centering is at $\widehat{\theta}$, not at c. The histogram of the bootstrap values in (9) substitutes for the appropriate t density in the real world.

Next, order the B values in (9) from smallest to largest. For the symmetric two-sided test we select the lower and upper $\alpha/2$ quantiles of the ordered values, t_l and t_u, respectively. Then the hypothesis is rejected if the value of t from (8) is less than t_l or if it is greater than t_u. The bootstrap p-value is simply the fraction of the absolute values from (9) which exceed the absolute value of t. One-sided testing is handled similarly.

Nonlinear function, $g(\beta)$. Suppose that we are interested in $g(\beta)$ rather than β where a function $g(\cdot) : \mathbb{R}^K \to \mathbb{R}^Q$ is a known function of β. The same logic applies to $g(\beta)$ and therefore bootstrapping provides a convenient way to compute the standard error, confidence interval, bias, and statistics for hypothesis testing for $g(\beta)$. This is because we can easily form the bootstrap samples $\{g(\beta^{(1)}), g(\beta^{(2)}), ..., g(\beta^{(B)})\}$ from $\{\beta^{(1)}, \beta^{(2)}, ..., \beta^{(B)}\}$.

The bootstrap can also fail ... as this example shows. Let $x_1, ..., x_n$ be independent draws from a uniform distribution on $(0, \theta)$. The usual estimator of θ is the largest order statistic, $\max(x_1, ..., x_n)$, which is the maximum likelihood estimator. Denote $\widehat{\theta} = \max(x_1, ..., x_n)$. The distribution function of $\widehat{\theta}$ is

$$P(\widehat{\theta} \leq y) = P(\text{all } x_i \leq y) = \left(\frac{y}{\theta}\right)^n,$$

and thus the density is

$$\begin{aligned} f_{\widehat{\theta}}(y) &= ny^{n-1}/\theta^n, \quad 0 \leq y \leq \theta, \\ &= 0, \quad \text{otherwise.} \end{aligned} \tag{10}$$

Suppose $\theta = 1$. If the bootstrap is effective, then the histogram of bootstrap estimates of $\theta(= 1)$ should approximate the density (10) with $\theta = 1$. However, none of these bootstrap estimates can exceed the largest of the n sample values. In fact, the probability that any one of the bootstrap estimates is equal to $\widehat{\theta} = \max(x_1, ..., x_n)$ is

$$1 - \left(\frac{n-1}{n}\right)^n = 1 - \left(1 - \frac{1}{n}\right)^n \approx 1 - e^{-1} = 0.632,$$

unless n is very small. Thus, the histogram of bootstrap estimates does not resemble the density (10). The distribution of the bootstrap estimates has a large probability mass at a single point, and the distribution of $\widehat{\theta}$ does not assign positive probability to any single point.

The jackknife. Elements of the jackknife were developed in a 1949 paper and a 1956 paper by the statistician M.H. Quenouille (1949, Approximate tests of correlation in time series, *Journal of the Royal Statistical Society Series B* **11** 68-84; 1956, Notes on bias in estimation, *Biometrika* **43** 353-360). The name "jackknife" was coined by John Tukey in unpublished work, and Tukey expanded Quenouille's ideas. The name Tukey chose expresses the role of the procedure as a rough-and-ready statistical tool. Part of the motivation for introduction of the jackknife is its function as a bias reducer. It can also be used to compute standard errors. However, the jackknife does not always work as hoped. For it to operate correctly, the estimator has to have local linear structure (R.G. Miller, 1974, The jackknife – a review, *Biometrika* **61** 1-15).

The jackknife is a resampling procedure that one can view as a forerunner to the bootstrap. As the previous paragraph suggests, it is less generally applicable than the bootstrap. But, it is often less computationally intensive than the bootstrap.

Given N data points, the jackknife uses N resamples, each of which leaves out just one data point. That is, the ith resample contains $N-1$ data points, all except the ith data point. One calculates the statistic of interest for each of the resamples (thus from $N-1$ data points in each resample), and then, to find the standard error of the statistic, one calculates the sample variance of the N values of the statistic (there is a scale adjustment at this stage). In the notation of the jackknife setting, the statistic for the ith resample is $g(\widehat{\beta}_{(-i)})$, $i = 1, ..., N$. Calculate the average of these N values of the statistic,

$$\overline{g}_N = \frac{1}{N}\sum_{i=1}^{N} g(\widehat{\beta}_{(-i)}).$$

Then the jackknife estimate of the standard error of the estimate $g(\widehat{\beta})$ is the square root of

$$\widehat{Var}_{jack} = \frac{N-1}{N}\sum_{i=1}^{N}\Big(g(\widehat{\beta}_{(-i)}) - \overline{g}_N\Big)^2. \tag{11}$$

There is a variation of this. Instead of deleting a single point for each resample, one can delete a group of points. For example if $N = km$, the original sample can be split into k groups of size m each, and for each resample one deletes one of the groups.

Efron and Stein (1981, The jackknife estimate of variance, *The Annals of Statistics* **9** 586-596) argue that the jackknife estimate (11) of the variance is biased upwards.

The same logic applies to the linear regression model by considering (y_i, x_i) as a data point.